

ACUTE CARDIOMETABOLIC RESPONSES TO A NOVEL TRAINING ROPE PROTOCOL IN CHILDREN

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ABSTRACT

Faigenbaum, AD, Kang, J, Ratamess, NA, Farrell, A, Golda, S, Stranieri, A, Coe, J, and Bush, JA. Acute cardiometabolic responses to a novel training rope protocol in children. *J Strength Cond Res* 32(5): 1197–1206, 2018—The purpose of this study was to quantify the acute cardiometabolic responses to a novel training rope (TR) protocol in children. Fifteen boys (10.6 ± 1.4 years) were tested for peak oxygen uptake ($\dot{V}O_2$) on a treadmill and subsequently (2–7 days later) performed a progressive 10-minute protocol of 5 TR exercises (EX): EX1: standing side-to-side waves, EX2: seated alternating waves, EX3: standing alternating waves, EX4: jumping jacks, and EX5: double-arm slams. Participants performed 2 sets of each TR exercise for 30 seconds per set with a 30-second rest interval between sets and exercises. A 12.8-m TR (4.1 kg) was used for all trials, and a metronome was used to standardize cadence. Peak values for $\dot{V}O_2$ and HR during the treadmill test were $47.4 \pm 8.8 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ and $195.1 \pm 6.6 \text{ b} \cdot \text{min}^{-1}$, respectively. During the TR protocol, mean $\dot{V}O_2$ and HR significantly increased ($p \leq 0.05$) from set 1 of EX1 ($10.3 \pm 2.6 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ and $103.2 \pm 11.6 \text{ b} \cdot \text{min}^{-1}$, respectively) to set 2 of EX5 ($30.0 \pm 3.9 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ and $168.6 \pm 11.8 \text{ b} \cdot \text{min}^{-1}$, respectively). Percentage of peak $\dot{V}O_2$ and peak HR values attained during the TR protocol ranged from 21.5 to 64.8% and from 52.9 to 86.4%, respectively. These descriptive data indicate that TR exercise can pose a moderate to vigorous cardiometabolic stimulus in children and with proper supervision and instruction may serve as a worthwhile complement to youth fitness programs.

KEY WORDS training rope, heart rate, oxygen consumption, children, resistance exercise, youth

INTRODUCTION

The efficacy and feasibility of strength and conditioning for children and adolescents have been reported in the literature and an increasing body of evidence has recognized the importance of resistance training as foundational for long-term athletic development (4,29). The benefits of integrating different types of resistance training into youth fitness programs have become particularly important in light of secular declines in measures of muscular fitness (i.e., muscular strength, muscular power, and local muscular endurance) (10,32) and motor skill performance (19) in modern day youth. For example, in the 1980s, the 50th percentile for modified pull-ups ranged from 6 to 10 repetitions, but the present 50th percentile for 6–9-year olds in the United States is 2–4 repetitions (27).

Recent reviews concluded that early exposure to developmentally appropriate strength and conditioning programs can improve markers of health, enhance physical literacy, and reduce injury risk in young athletes (38,40). Such reports have led to efforts to further investigate the neuromuscular effects of different types of resistance training on children (11,13,22). However, data on the cardiometabolic responses to resistance exercise in youth are limited (20,37). Although previous pediatric research has examined the physiological responses to short sprint bursts (24) and supramaximal bouts of cycle ergometry (9), additional investigations examining the cardiovascular and metabolic responses to resistance exercise in children are needed to fill this research gap.

A variety of exercise modalities including free weights, weight machines, elastic bands, and medicine balls have been used in youth strength and conditioning programs (11,13,22). More recently, practitioners have started to incorporate training rope (TR) protocols into physical education classes (7). Unlike jump ropes, which are thin and light, a TR is thicker, heavier, and longer and can be used for multiple purposes including pulling, slamming, and wave training (6). The available data indicate that TR protocols can be effective conditioning tools for adults because of the potent cardiovascular and metabolic demands of this type of exercise (17,34,35). Ratamess et al. (34) quantified the acute cardiometabolic responses to 13 different resistance exercise protocols in resistance-trained men and found that the TR protocol elicited the largest acute cardiometabolic responses

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compared with all other traditional resistance training and body weight exercises. Others have shown that TR protocols provide a vigorous intensity cardiometabolic stimulus in adults as evidenced by very high peak heart rates (HR) (94% of age-predicted maximum) and energy expenditure per unit of time ($41 \text{ kJ} \cdot \text{min}^{-1}$) (17).

At present, there is a strong rationale for the identification of novel and time-effective exercise interventions that can modulate disease risk factors and improve health outcomes in children and adolescents (5,12). Notably, there is growing interest in high-intensity interval training because short bouts of vigorous physical activity (e.g., running-based sessions $>90\%$ HR maximum) have been found to be favorably related to health outcomes in youth (12). Of particular importance to the current investigation, no published reports have specifically examined the acute cardiometabolic responses to a novel TR protocol in children. It is important to understand the acute cardiometabolic responses to novel resistance exercise modalities in youth because the life-changing consequences of low levels of muscular fitness are not limited to adult populations (15). Further examination of the acute responses to different modes of resistance exercise is necessary to better understand the basic mechanisms underpinning training-induced adaptations. The results of this investigation could also enhance the design of school- and community-based youth fitness programs. Accordingly, the purpose of this descriptive study was to quantify the acute cardiometabolic responses to a novel TR protocol in children. It was hypothesized that TR exercises would elicit a significant increase in the cardiometabolic demand, in line with previous observations from research involving children and adults (13,34).

METHODS

Experimental Approach to the Problem

To examine the primary hypothesis of this investigation, participants were tested for $\dot{V}O_{2\text{peak}}$ and subsequently performed a 10-minute TR protocol, which was developmentally appropriate for children. The TR protocol consisted of 5 progressive exercises of increasing complexity and participants performed 2 sets of each TR exercise for 30 seconds per set with a 30-second rest interval between sets and exercises. Breath-by-breath oxygen consumption, HR, and ratings of perceived exertion (RPE) data were collected. This descriptive experimental design allowed us to quantify the cardiometabolic responses to an acute bout of TR exercise in children.

Subjects

Fifteen boys between 7.7 and 11.9 years of age ($\pm SD$ mean age 10.6 ± 1.4 years, height 142.5 ± 8.4 ; body mass 37.4 ± 8.0 kg) from local sport teams (primarily basketball, lacrosse, and soccer) volunteered to participate in this study. A modified physical activity readiness questionnaire (PAR-Q) was used to evaluate the health status of participants and assess

the safety for performing maximal exercise. Exclusion criteria were the following: use of medication affecting exercise capacity; cardiopulmonary or metabolic disease; orthopedic limitation; or positive responses from parents to one or more of the PAR-Q questions pertaining to their child's health. No participant was excluded from participation. This study was approved by the Institutional Review Board at The College of New Jersey. All parents signed a parental permission form, and all participants signed a child assent form and were informed of the benefits and risks of this investigation.

Procedures

Maximal Aerobic Capacity Testing. All participants reported to the Human Performance Laboratory during the spring semester (February–April) at a standardized time of day at least 2 hours postprandial for maximal aerobic capacity testing. Participants were asked to refrain from vigorous exercise (e.g., sports competition) for at least 24 hours before the testing session. $\dot{V}O_{2\text{peak}}$ was assessed using the Fitkids treadmill test protocol (28) and a metabolic system (MedGraphics ULTIMA Metabolic System; MedGraphics Corporation, St Paul, MN, USA). Gas analyzers were calibrated before each trial using gases provided by MedGraphics Corporation: (a) calibration gas: 5% CO_2 , 12% O_2 , and balance N_2 and (b) reference gas: 21% O_2 and balance N_2 . Each participant was fitted with a child-size respiratory mask that was placed over the participants face, fastened, and carefully checked for proper sealing. Before the test, each participant sat quietly in a chair for 5 minutes to collect baseline data. The Fitkids treadmill test protocol consisted of a 90-second warm-up period ($3.5 \text{ km} \cdot \text{h}^{-1}$, 0% grade) followed by the initiation of the test at $3.5 \text{ km} \cdot \text{h}^{-1}$ and a 1% grade for 90 seconds followed by incremental increases in both speed ($0.5 \text{ km} \cdot \text{h}^{-1}$) and incline (2%) every 90 seconds until an incline of 15% was attained (25). After this stage, the incline was kept constant at 15%, and incremental increases of speed ($0.5 \text{ km} \cdot \text{h}^{-1}$) were performed every 90 seconds. All participants were verbally encouraged to continue exercising until volitional exhaustion.

Breath-by-breath $\dot{V}O_2$ data were obtained, and $\dot{V}O_{2\text{peak}}$ was determined by recording the highest measure observed during the test (2). Peak HR was defined as the highest value achieved during the test. HR was monitored using a soft chest strap with an HR sensor (Model A300; Polar Electro Inc, Woodbury, NY, USA). After the test, HR data were downloaded using a computer software program for analysis. The treadmill test was deemed maximal when at least one of the following objective criteria was met: HR peak $>180 \text{ b} \cdot \text{min}^{-1}$ or a respiratory exchange ratio (RER) > 1.0 (3,18). Subjective criteria for maximal effort (e.g., unsteady running, facial flushing, and clear unwillingness to continue despite verbal encouragement) were also monitored. During the test, participants were asked to manually signal without verbalizing their RPE on a visually presented scale consisting of verbal expressions along with a numerical response range of 0–10 and 5 pictorial descriptors that represent a child at

TABLE 1. Training rope (TR) exercise protocol.

Time (min)	Set	TR exercise	Repetitions	Rest interval (s)
0–0.5	1	Standing side-to-side wave	30	30
1.0–1.5	2		30	30
2.0–2.5	3	Seated alternating waves	60	30
3.0–3.5	4		60	30
4.0–4.5	5	Standing alternating waves	60	30
5.0–5.5	6		60	30
6.0–6.5	7	Jumping jacks	20	30
7.0–7.5	8		20	30
8.0–8.5	9	Double-arm slams	20	30
9.0–9.5	10		20	30
9.5–10				30

varying levels of exertion (16). After the test, participants were monitored for 5 minutes while walking to ensure normal recovery of HR ($2.0 \text{ km} \cdot \text{h}^{-1}$, 0% grade).

Training Rope Protocol. A standard nylon TR (mass 4.1 kg, length 12.8 m; diameter 2.5 cm, Perform Better, Providence, RI, USA) was used for all study procedures. The TR was anchored around 2 handles ($\sim 30 \text{ cm}$ apart) of a heavy sand-

bag placed on the floor. Pilot work before the study found that this anchoring pattern minimized rope collisions and best standardized TR exercise performance in children. The TR protocol consisted of the following 5 exercises: (a) standing side-to-side waves (EX1), (b) seated alternating waves (EX2), (c) standing alternating waves (EX3), (d) jumping jacks (EX4), and (e) double-arm slams (EX5) (Table 1). A photograph of each TR exercise is shown in Figures 1 and 2, and descriptions of TR exercises are available elsewhere (6,39). The 5 TR exercises were performed in successive order

with each exercise set lasting 30 seconds. Each exercise was performed for 2 sets with a rest interval of 30 seconds in between sets and exercises. The total duration of the entire protocol lasted 10 minutes (including 30-second recovery after the last exercise).

Participants were asked to follow a specific cadence using a metronome and verbal cues to complete a target number of repetitions during each 30-second set. For

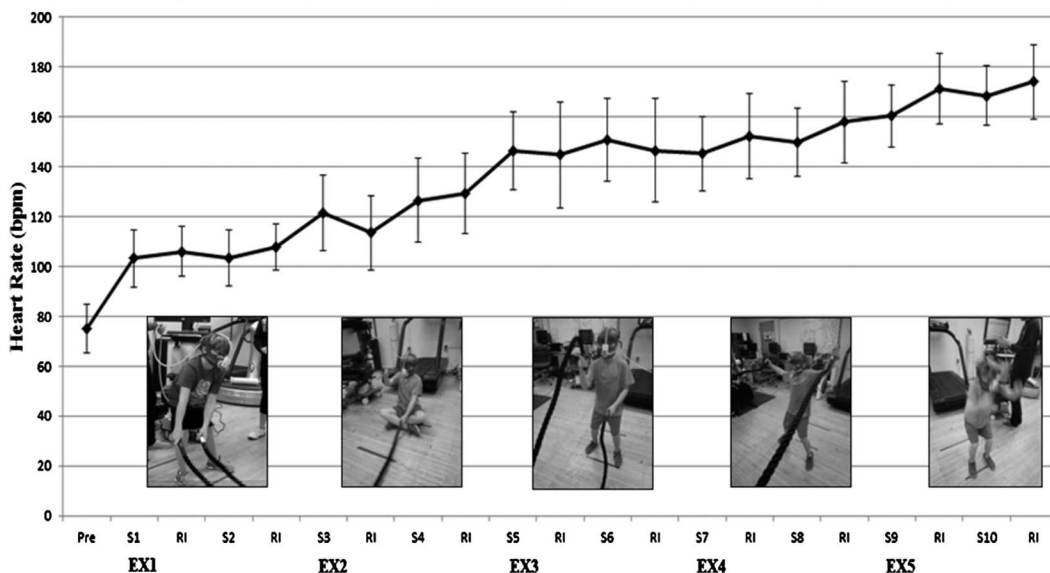


Figure 1. Acute heart rate (HR) responses (mean $\pm$ SD) during the training rope protocol. PRE = baseline; EX = exercise; S = set 1; RI = rest interval. See Table 3 for significant differences between sets.

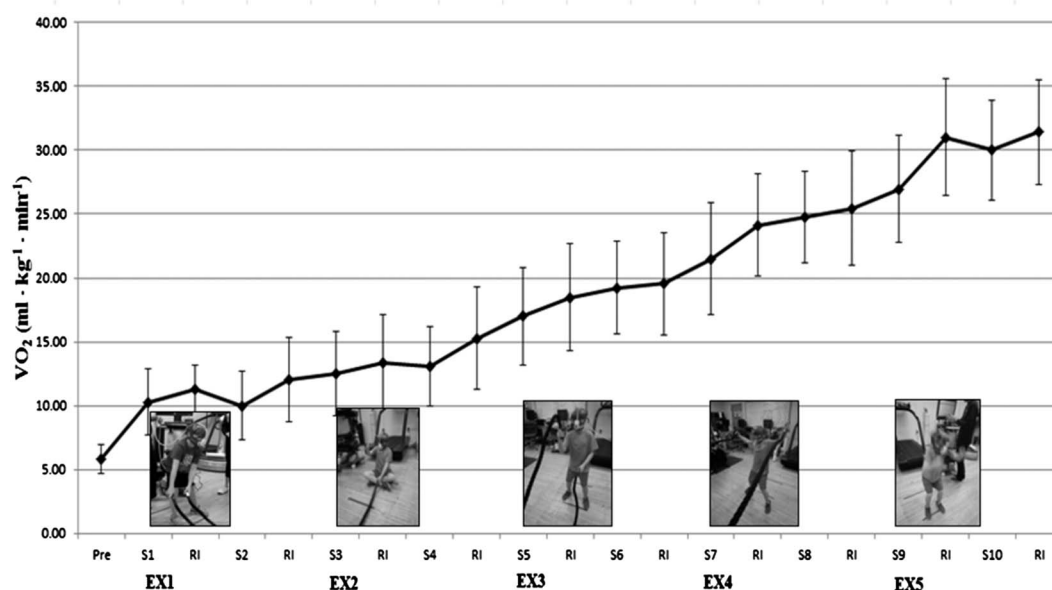


Figure 2. Acute relative oxygen uptake responses (mean $\pm$ SD) during the training rope protocol. PRE = baseline; EX = exercise; S = set; RI = rest interval. See Table 3 for significant differences between sets.

alternating wave exercises (EX2 and EX3), participants attempted to perform 30 repetitions for each arm totaling 60 repetitions per set. The TR protocol was recorded on video, and repetitions for each set were subsequently counted and analyzed. Each TR exercise was associated with a child-friendly coaching cue that was provided to the participants to assist them in maintaining proper technique. For the 5 TR exercises, their respective coaching cues were EX1: make a snake, EX2: play the drums, EX3: double Dutch, EX4: scarecrow, and EX5: pancakes.

Participants became familiar with the TR protocol and study procedures during a 30-minute familiarization session. A Certified Strength and Conditioning Specialist demonstrated the proper TR technique, and participants received instructional cues and constructive feedback on the quality of each movement. The learning process was reinforced through a direct instructional model as participants gained movement skill competency. The familiarization session focused on proper movement patterns and the ability to perform different TR exercises correctly at the desired cadence. Of note, the TR protocol was designed to be feasible and developmentally appropriate for children at a novice level; therefore, the complexity of each TR exercise increased in a progressive manner from relatively simple movements to more challenging exercises. Participants also had an opportunity to perform TR exercises while wearing a respiratory mask. Before the familiarization session, height was

measured to the nearest 0.1 cm using a wall-mounted stadiometer, and body mass was measured to the nearest 0.5 kg using an electronic scale. For both measurements, participants wore light cloths and no shoes.

Training Rope Experimental Trial. Participants returned to the Human Performance Laboratory at least 2 hours post-prandial within 2–7 days of the maximal aerobic capacity test to perform the TR test protocol. On arrival, each participant was asked to drink water ad libitum to prehydrate and was fitted with a child-size respiratory mask that was placed over the participant's face, fastened, and carefully checked for proper sealing. Each participant was fitted with a Polar HR monitor (model A300; Polar Electro Inc, Woodbury, NY, USA) that was used to measure HR before, during, and after the TR protocol. HR data were downloaded for analysis using a computer software program. Breath-by-breath $\dot{V}O_2$ was measured during the TR protocol using a metabolic system (MedGraphics ULTIMA Metabolic System; MedGraphics Corporation). HR data collected were the mean protocol values collected immediately after each set. Participants were asked to manually indicate their RPE after each set using the same 0–10 RPE scale used during maximal aerobic capacity testing (16).

Before the TR protocol, each participant sat quietly in a chair for 5 minutes to collect baseline data. Subsequently, each participant performed about 3–4 minutes of dynamic stretching (e.g., arm circles and knee lifts) and lightweight (1 kg) medicine ball exercises (e.g., twisting and knee bends).

TABLE 2. Comparison of mean oxygen uptake ($\dot{V}O_2$), minute ventilation (V_E), RER, HR, and RPE during training rope exercise (EX).*†

EX	Set (S)	$\dot{V}O_2$ (L·min ⁻¹)	$\dot{V}O_2$ (ml·kg ⁻¹ ·min ⁻¹)	V_E (L·min ⁻¹)	RER	HR b·min ⁻¹	RPE
EX1	S1	0.39 ± 0.12‡§	10.3 ± 2.6‡§	10.3 ± 2.9‡§	0.84 ± 0.1‡§	103.2 ± 11.6‡§	0.5 ± 0.7‡§
	S2	0.38 ± 0.12‡§	10.0 ± 2.7 ‡§	10.2 ± 3.2‡§	0.84 ± 0.1‡§	103.5 ± 11.4‡§	0.9 ± 1.1‡§
EX2	S3	0.47 ± 0.15 ‡§	12.5 ± 3.3‡§	16.0 ± 5.7‡§	0.96 ± 0.1‡§	121.6 ± 15.2‡§	2.3 ± 1.2‡§
	S4	0.49 ± 0.12 ‡§	13.1 ± 3.1‡§	20.6 ± 7.7‡§	1.18 ± 0.2	126.5 ± 16.9‡§	3.5 ± 1.6‡§
EX3	S5	0.63 ± 0.14‡§	17.0 ± 3.8‡§	26.8 ± 8.5‡§	1.17 ± 0.2	146.5 ± 15.5‡§	4.3 ± 1.7‡§
	S6	0.71 ± 0.13‡§	19.3 ± 3.6‡§	29.8 ± 8.6‡§	1.15 ± 0.2	150.7 ± 16.7‡§	5.2 ± 1.8‡§
EX4	S7	0.80 ± 0.18‡§	21.5 ± 4.4‡§	27.9 ± 6.9‡§	1.03 ± 0.1‡§	145.2 ± 15.0‡§	5.1 ± 1.7‡§
	S8	0.93 ± 0.20‡§	24.8 ± 3.6‡§	31.6 ± 7.2‡§	1.01 ± 0.1‡§	149.8 ± 13.5‡§	5.7 ± 2.0‡§
EX5	S9	1.01 ± 0.22‡§	27.0 ± 4.2‡§	34.9 ± 6.8‡§	1.04 ± 0.8§§	160.5 ± 12.4‡§	6.5 ± 1.8‡§
	S10	1.12 ± 0.19‡§	30.0 ± 3.9‡§	40.8 ± 6.7‡§	1.07 ± 0.1‡§	168.6 ± 11.8‡§	7.3 ± 2.0‡§

*RER = respiratory exchange ratio; HR = heart rate; RPE = ratings of perceived exertion.

†Description of each EX is provided in the Methods section.

‡ $p \leq 0.05$ vs. S5.

§ $p \leq 0.05$ vs. S6.

|| $p \leq 0.05$ vs. S7.

|| $p \leq 0.05$ vs. S8.

$p \leq 0.05$ vs. S9.

** $p \leq 0.05$ vs. S10.

‡† $p \leq 0.05$ vs. S3.

‡‡ $p \leq 0.05$ vs. S4.

§§ $p \leq 0.05$ vs. S1.

|||| $p \leq 0.05$ vs. S2.

TABLE 3. Comparison of relative oxygen uptake (% $\dot{V}O_{2peak}$) and relative heart rate (% HR peak) during training rope exercises (EX).*

EX	Set (S)	% $\dot{V}O_{2peak}$	%HRpeak
EX1	S1	22.0 $\pm$ 5.5	52.9 $\pm$ 5.2
	S2	21.5 $\pm$ 5.7	53.0 $\pm$ 5.1
EX2	S3	26.6 $\pm$ 5.8	62.3 $\pm$ 7.2
	S4	30.0 $\pm$ 5.8	58.2 $\pm$ 7.4
EX3	S5	36.5 $\pm$ 7.5	75.1 $\pm$ 6.6
	S6	41.2 $\pm$ 7.2	77.2 $\pm$ 7.5
EX4	S7	46.3 $\pm$ 10.0	74.4 $\pm$ 6.6
	S8	53.5 $\pm$ 9.9	76.7 $\pm$ 5.6
EX5	S9	57.9 $\pm$ 9.8	82.2 $\pm$ 4.8
	S10	64.8 $\pm$ 10.9	86.4 $\pm$ 4.3

*Description of each EX is provided in the Methods section.

Once complete, an approximate 1-minute period ensued in which the researcher reviewed session instructions and the participant assumed the starting position for the first exercise. A research assistant video-recorded and monitored the number of rope oscillations during each time interval and subsequently counted and analyzed the number of repetitions. During the TR protocol, a research assistant periodically checked the facemask for proper fit. Participants began in an athletic stance (except EX2, which began in the seated position) and were instructed to perform each TR exercise at a specific cadence with the proper exercise technique. During each rest interval, a research assistant provided a quick review of the upcoming TR exercise and reinforced exercise-specific coaching cues to maintain proper technique. The 10-minute TR exercise protocol consisted of 2 sets of 30 seconds on 5 different TR exercises with a 30-second rest interval between sets and exercises as outlined in Table 1. Verbal encouragement was used consistently throughout the TR trials. Tape markers were placed on the floor to ensure proper foot position. A metronome was used to standardize cadence. All participants performed the same TR exercises in the same order. The design of the TR protocol was based on previous school-based interventions, which included short bouts of exercise with 30 seconds of rest between sets and exercises (13,14).

Values for HR, absolute $\dot{V}O_2$, relative $\dot{V}O_2$, minute ventilation (V_E), and RER were recorded during the entire TR protocol. Individual breath-by-breath data points for all metabolic variables were averaged for the entire set of each TR exercise. The time corresponding to the initiation of each set and the rest interval length between each set were carefully monitored and labeled during each protocol. The values between these time points were subsequently averaged and analyzed.

Statistical Analyses

Descriptive statistics (mean $\pm$ SD) were calculated for all dependent variables. A 1 (group) $\times$ 10 (sets) analysis of variance with repeated measures was used to analyze within-participant cardiometabolic and RPE data. Subsequently, Tukey's post hoc tests were used to determine differences when significant main effects were obtained. For all statistical tests, a probability level of $p \leq 0.05$ denoted statistical significance. Statistical analyses were conducted in SPSS (version 18.0; SPSS, Chicago, IL).

RESULTS

All participants successfully completed all study-related procedures, and no injuries or unexpected events occurred during testing trials. Participants were healthy boys with a maximal treadmill test peak $\dot{V}O_2$ of 47.4 ± 8.8 ml $\cdot$ kg $^{-1}\cdot$ min $^{-1}$ and a peak HR of 195.1 ± 6.6 b $\cdot$ min $^{-1}$. A comparison with age-related normative values for the Fitkids treadmill test indicates that the mean time to exhaustion of 10.5 ± 1.5 minutes for participants in the current investigation was about the 50th percentile (26). Cardiometabolic and perceptual responses are presented in Table 2. Significant ($p \leq 0.05$) effects of TR exercise were observed for all variables, and there was a gradual increase in these parameters throughout the 10 sets (S) of the TR protocol. Our post hoc comparisons revealed a progressive increase in cardiometabolic demand as $\dot{V}O_2$, V_E , and HR increased significantly ($p \leq 0.05$) from EX1 (standing side-to-side wave) to EX5 (double-arm slams) with the highest peak $\dot{V}O_2$, V_E , and HR reaching 36.7 ± 4.5 ml $\cdot$ kg $^{-1}\cdot$ min $^{-1}$, 51.2 ± 8.1 L $\cdot$ min $^{-1}$, and 180.5 ± 12.5 b $\cdot$ min $^{-1}$, respectively, during EX5. The highest mean values for $\dot{V}O_2$, V_E , and HR were 30 ± 3.9 ml $\cdot$ kg $^{-1}\cdot$ min $^{-1}$, 40.8 ± 6.7 L $\cdot$ min $^{-1}$, and 168.6 ± 11.8 b $\cdot$ min $^{-1}$, respectively, during EX5.

Analysis of $\dot{V}O_2$ during each set (S) of the TR protocol revealed that S10 was significantly higher than S1–S9; S9 was significantly higher than S1–S7; S8 was significantly higher than S1–S7; S7 was significantly higher than S1–S5; S6 was significantly higher than S1–S4; S5 was significantly higher than S1–S4; and S4 was significantly higher than S2. Values for V_E and HR increased with each successive exercise and tended to parallel $\dot{V}O_2$ data. Figures 1 and 2 depict the gradual and progressive increase in HR and relative $\dot{V}O_2$, respectively, during the TR protocol. Percentage of peak $\dot{V}O_2$ and peak HR values attained during the TR protocol ranged from 21.5 to 64.8% and from 52.9 to 86.4%, respectively (Table 3). The RER increased from 0.84 ± 0.1 during S1 to 1.07 ± 0.1 during S10.

Significant ($p \leq 0.05$) effects of TR exercises were observed for RPE. There was a gradual increase in perceptual responses from EX1 through EX5 (Table 2). Our post hoc comparisons revealed a progressive increase in RPE during the TR protocol as RPE increased significantly

($p \leq 0.05$) between successive exercises with the highest RPE reaching 7.3 ± 2.0 (of 10) during the second set of EX5. The progressive increase in RPE during the TR protocol mirrored increases in cardiometabolic responses.

DISCUSSION

The aim of this study was to examine the acute cardiometabolic responses to a novel TR protocol in children. In line with our hypothesis, our findings demonstrate that a 10-minute bout of 5 TR exercises comprising 30-second work sets interspersed with 30 seconds of passive recovery can pose a potent cardiovascular and metabolic stimulus in children. Although the cardiovascular and metabolic responses to aerobic-based sprint training (9,24) and different types of resistance training (20,37) have been examined in youth, our novel findings highlight the potential cardiometabolic benefits of TR exercise.

Although there are a wide variety of TR exercises that can be performed, our TR protocol was purposely designed for children with no or limited TR experience. Our findings indicate that a series of 5 TR exercises can place a moderate to vigorous cardiometabolic demand on children as evidenced by mean HR and $\dot{V}O_2$ values reaching 168.6 ± 11.8 $b \cdot min^{-1}$ (86.4% HR_{peak}) and 30.0 ± 3.9 $ml \cdot kg^{-1} \cdot min^{-1}$ (64.8% $\dot{V}O_{2peak}$), respectively (Tables 2 and 3). Peak HR and $\dot{V}O_2$ values during the TR protocol increased from 111.4 ± 11.3 $b \cdot min^{-1}$ and 11.5 ± 3.3 $ml \cdot kg^{-1} \cdot min^{-1}$, respectively, during EX1 to 180.5 ± 12.5 $b \cdot min^{-1}$ and 36.7 ± 4.5 $ml \cdot kg^{-1} \cdot min^{-1}$, respectively, during EX5. Compared with a standard classification of physical activity intensity, the HR response to all 5 BR exercises was within the moderate to vigorous HR zone (1). Our HR data from EX5 are consistent with findings of Harris et al. (20), who reported mean HR of 170 ± 9.1 and 179 ± 5.6 $b \cdot min^{-1}$ after an acute bout of multiset resistance training and high-intensity interval training, respectively, in 12–13-year-old youth. Although HR data from TR exercise are limited, others reported an HR of about 80–90% maximum during bouts of TR exercise in adults (17,35).

During the TR protocol, oxygen consumption progressively increased from 10.3 ± 2.6 $ml \cdot kg^{-1} \cdot min^{-1}$ (22% $\dot{V}O_{2peak}$) during EX1 (standing side-to-side wave) to 30.0 ± 3.9 $ml \cdot kg^{-1} \cdot min^{-1}$ (64.8% $\dot{V}O_{2peak}$) during EX5 (double-arm slams). These findings are comparable to other investigations that reported $\dot{V}O_2$ values of 24.9 ± 3.2 $ml \cdot kg^{-1} \cdot min^{-1}$ and 33.8 ± 5.2 $ml \cdot kg^{-1} \cdot min^{-1}$ during intermittent bouts of resistance training and high-intensity interval training, respectively, in early adolescents (20). Our observations are consistent with findings of Ratamess et al. (34), who measured the acute metabolic responses to 13 different resistance exercise protocols in adults and found that the mean $\dot{V}O_2$ of 24.6 $ml \cdot kg^{-1} \cdot min^{-1}$ (50.8% $\dot{V}O_{2max}$) was highest during the TR protocol than with traditional resistance (e.g., squat and bench press) and body weight (e.g., burpee and push-up) exercises. In the aforementioned

study on adults, the BR protocol used a 10.9-kg TR and consisted of 3 sets of 30-second bouts of exercise (performed with self-selected maximal cadence) with a 2-minute rest interval between sets, with each set consisting of 10-second bouts of single-arm alternating waves, double-arm waves with a half squat followed by double-arm rope slams with a half squat (34). In the present investigation, a 4.1-kg TR was used to perform 2 sets of 30 seconds on 5 different TR exercises with a 30-second rest interval between sets and exercises.

It was not surprising that the relative oxygen consumption increased as the TR protocol progressed from standing side-to-side waves (EX1) to double-arms slams (EX5). This increase may have been due to the cumulative effects of fatigue on subsequent exercise performance, cardiovascular drift, and the greater complexity or perceived intensity of the TR exercises selected toward the end of the protocol (35). As previously observed in adults (34), the double-arm slam is perceived to be a more intense exercise because it has a substantially lower body contribution that requires participants to forcibly slam the ropes against the ground at near maximal or maximal velocity during each repetition. In support of these observations, Fountaine et al. (17) quantified the cardiometabolic cost of a 10-minute bout of TR exercise (15 seconds of double-arm slams and 45 seconds rest) using a 16.3-kg TR in adult men and women and reported a peak $\dot{V}O_2$ of 35.4 $ml \cdot kg^{-1} \cdot min^{-1}$ and peak HR of 178 ± 11 $b \cdot min^{-1}$, which was 94% of age-predicted maximum.

Since movement complexity, intensity, and muscle mass involvement drive acute oxygen consumption as observed during high-intensity intermittent exercise (33), it seems that complex and intense TR exercises that require large muscle mass activation can pose a stronger cardiometabolic stimulus than side-to-side waves or alternating waves, which require less motion of the hips, knees, and ankles. It is important to note that the major aim of this study was to quantify 1 novel TR protocol and not to directly compare each of the 5 TR exercises. Thus, one must view our results within the context of the nonrandomized sequence by which the exercises were performed and must understand that the cardiometabolic responses to each exercise were likely affected by the sequence (and subsequent fatigue induced by the previous exercise). However, our findings indicate that selected TR exercises could be an effective means of increasing gross energy expenditure when using a protocol similar to the one used in this study. In our investigation, energy expenditure progressively increased from 1.91 $kcal \cdot min^{-1}$ during EX1 to 5.6 $kcal \cdot min^{-1}$ during EX5.

Our findings indicate that the choice of TR exercise is a primary determinant of the cardiometabolic responses to this type of training, although other program design variables including the length and mass of the TR, repetition velocity, and rest interval length in between sets and exercises should also be considered (34,35). In our investigation, EX1 (standing side-to-side wave) was perceived to be

less complex than EX5 (double-arm slams). Although the cumulative effects of fatigue should be considered when evaluating our data from a single 10-minute session of 5 TR exercises, the intensity, cadence, rest intervals, and amount of muscle mass recruited during each exercise have been found to be primary contributors to the acute metabolic responses in adults (23,33,35). Additionally, participants attempted to complete 20 repetitions of the double-arm slams within each 30-second interval. Video analysis revealed that while 20 repetitions was a target, participants actually performed 21.9 ± 2.3 and 21.5 ± 2.0 repetitions during the first and second set, respectively, of EX5. The greater effort required to perform double-arm slams and the faster cadence likely increased the mean cardiometabolic responses. Of interest, participants were also able to achieve the desired cadence during both sets of EX1 (35.2 ± 4.0 and 36.1 ± 1.0 repetitions) and EX4 (20.4 ± 2.5 and 20.6 ± 2.5 repetitions), but the larger variability in cadence performance during EX2 (64.3 ± 15.0 and 62.3 ± 12.6 repetitions) and EX3 (57.9 ± 16.4 and 55.4 ± 14.2 repetitions) that required alternating wave movements in the seated and standing positions is noteworthy. The coordination required to perform alternating waves and different muscle activity during unilateral bilateral TR exercises may explain these observations (8).

Our cardiometabolic data are consistent with the participant's subjective RPE, which progressively increased from 0.5 to 0.9/10 ("very, very, easy") after EX1 to 6.5–7.3/10 "Hard" or "Very hard" after EX5. RPE scales have been found to be useful and practical measures for assessing resistance exercise intensity in youth (16,36). For example, Robertson et al. (36) reported that children could concurrently and differently rate their perceived intensity of muscle hurt and exertion during upper and lower body resistance exercise. Others reported that RPE could be used to quantify the interval training load in youth taekwondo training sessions (30). Collectively, these findings support the use of RPE to monitor TR exercise intensity in children. Because of the novelty, unique demands, and physical requirements of TR exercises, RPE may be particularly useful when subjectively assessing BR performance in beginners. Of potential relevance, others noted that a single bout of high-intensity interval exercise was associated with greater postexercise enjoyment than a bout of moderate-intensity interval exercise in adolescents (31).

Our findings extend the current literature base and demonstrate exercise-specific cardiometabolic responses to TR exercise in children. The structure of the technique-driven TR protocol in this investigation allowed all participants to complete 2 sets of 5 exercises that gradually progressed in exercise intensity. The 30-second rest interval between sets and exercises seemed to be sufficient and consistent with previous investigations involving children (13,14). It seems that TR exercise may be a useful addition to supervised youth strength and conditioning programs because the complexity, intensity, cadence, and rest interval between sets and exercises can influ-

ence the cardiometabolic response and consequent training-induced adaptations. We also found that TR exercises can be a challenging and enjoyable method of exercise for youth as evidenced by 100% compliance with research instructions and testing protocols. However, it must be underscored that all procedures were closely supervised by qualified professionals, and an attempt was made to make the TR protocol challenging yet enjoyable with "child friendly" coaching cues and enthusiastic instruction from research assistants.

Despite the strengths of this study, there are some limitations. The participants in our study included healthy boys, so the homogeneity of the sample limits generalizability of our results to other populations including those who are less physically active. The maturation status of the participants was not assessed, and this might have influenced the results. It is also important to consider the acute program variables (e.g., choice and order of exercises) and size of the TR used in our study because these factors will impact the cardiometabolic demands of TR exercise. Additionally, our data are from one 10-minute bout of 5 different TR exercises performed successively, and therefore, the cumulative effects of fatigue and cardiovascular drift should be considered when interpreting the findings. However, the TR protocol in this study was purposely designed for children with no or limited TR experience, and it is unlikely, based on previous investigations (13,14) that children would be able to complete a 10-minute TR protocol of total body exercises such as double-arm slams. Further studies accounting for these limitations are needed to isolate the independent cardiometabolic responses to specific TR exercises in a more diverse sample of participants characterized by different health and fitness levels.

PRACTICAL APPLICATIONS

Historically, ropes were used in physical education for different purposes including climbing and pulling. Nowadays, an increasing number of children and adults seem to be using TR in fitness centers and sport facilities. Our findings indicate that exercising with a TR could be a worthwhile addition to school- and community-based youth programs because this type of exercise has the potential to target several training goals including enhanced neuromuscular and cardiorespiratory fitness. For example, TR could be integrated into a developmentally appropriate fitness circuit that provides children with an opportunity to enhance muscular fitness, improve motor skill performance, and gain confidence in their abilities to be physically active (7). With qualified instruction and supervision by professionals who understand the principles of pediatric exercise science and resistance training, children can learn health-promotion concepts and exercises while participating in a program that includes variety, progression, and proper rest intervals.

Because moderate to vigorous physical activity levels during physical education classes and youth sports practice are falling short of expectations (21,28), the potential value of

incorporating TR into exercise programs and training sessions should be considered. Notably, the high cardiometabolic demand of TR exercise could potentially provide a novel stimulus for improving aerobic fitness and cardiovascular disease biomarkers in youth (12). However, the importance of qualified supervision and technique-driven progression must be underscored to reduce the risk of physical activity-related injuries, maintain adherence, and promote enjoyment of the exercise experience. Additional research is needed to expand these observations and improve our understanding of program design considerations that optimize training-induced adaptations across a range of pediatric groups.

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